

## Chuqur o'rganish usullari

### 1-mavzu. Sun'iy neyron, perceptron va logistik neyron — Practice

**Mavzu:** sun'iy neyron  $a = \varphi(\mathbf{w}^T \mathbf{x} + b)$ , faollashtirish funksiyalari (sigmoid, tanh, ReLU, Leaky ReLU, softplus), perceptron va Novikoff teoremasi, XOR va yashirin qatlam, logistik neyron, binary cross-entropy, gradient tushish (gradient descent), boshlang'ich qiymatlar (initialization).

**Maqsad:** (1) neyron va qatlam hisobini qo'lda bajarish; (2) asosiy tasdiqlarni isbotlash; (3) NumPy va PyTorch'da neyron, perceptron va kichik MLP ni yozish hamda natijani tahlil qilish.

**Baholash:** har bir topshiriq 1 ball, jami 30 ball. Hisob topshiriqlarida javob 4 ta o'nlik xona aniqlikda; kod topshiriqlarida (17–30) kod xatosiz ishlashi, chop etilgan natija ko'rsatilgan formatda bo'lishi va 1–2 jumlati xulosa talab qilinadi.

**Muhit:** Python 3.11/3.12, `pip install -r requirements.txt` (versiyalar qat'iy: numpy 2.4.4, scipy 1.17.1, scikit-learn 1.8.0, torch 2.14.0, faqat CPU). Barcha ma'lumotlar kutubxona ichida (`sklearn.datasets`) — internet shart emas. Kod topshiriqlari bitta `solutions.py` faylida `task_17()`, ..., `task_30()` funksiyalari sifatida yoziladi. Hujjatlar: <https://pytorch.org/docs/stable/nn.html>, <https://numpy.org/doc/stable/reference/>, <https://scikit-learn.org/stable/api/index.html>.

#### A. Qo'lda hisoblash, isbot va tushunchalar (1–16)

- (hisob) Neyron:  $\mathbf{x} = (1; -2; 3)$ ,  $\mathbf{w} = (0,4; 0,1; -0,2)$ ,  $b = 0,3$ .  $z = \mathbf{w}^T \mathbf{x} + b$  ni, so'ng sigmoid, tanh, ReLU va Leaky ReLU ( $\alpha = 0,1$ ) chiqishlarini hamda sigmoid, tanh va ReLU ning shu  $z$  dagi hosilalarini toping. Neyronda nechta o'rganiladigan parametr bor?
- (hisob) Paket  $X = \begin{pmatrix} 1 & 0 \\ 2 & -1 \\ 0 & 3 \end{pmatrix}$ , qatlam  $W = \begin{pmatrix} 1 & -1 & 0,5 \\ 2 & 0 & -1 \end{pmatrix}$ ,  $\mathbf{b} = (0; 1; -0,5)$ .  $Z = XW + \mathbf{1b}^T$  va  $A = \text{ReLU}(Z)$  ni toping. Qatlam parametrlari soni nechta? `load_digits` tasvirlari ( $d = 64$ ) uchun  $m = 32$  neyronli qatlamda-chi?
- (hisob) Giperyuza  $2x_1 - x_2 + 2x_3 - 3 = 0$ .  $A = (2; 0; 1)$  va  $B = (0; 1; 1)$  nuqtalar uchun  $z$  qiymatini, giperyuzagacha masofani va qaysi yarim fazoda yotishini aniqlang.  $A$  ning giperyuzadagi ortogonal proyeksiyasini toping.
- (isbot)  $\tanh u = 2\sigma(2u) - 1$  ayniyatidan foydalanib, bir yashirin qatlamli tanh-tarmoq  $f(\mathbf{x}) = \mathbf{v}^T \tanh(W\mathbf{x} + \mathbf{b}) + c$  sigmoid-tarmoqqa  $W' = 2W$ ,  $\mathbf{b}' = 2\mathbf{b}$ ,  $\mathbf{v}' = 2\mathbf{v}$ ,  $c' = c - \sum_j v_j$  parametrlar bilan aynan teng ekanini isbotlang. Tekshiring:  $W = \begin{pmatrix} 1 & -1 \\ 0,5 & 2 \end{pmatrix}$ ,  $\mathbf{b} = (0; -1)$ ,  $\mathbf{v} = (1; -2)$ ,  $c = 0,5$ ,  $\mathbf{x} = (1; 1)$  da ikkala ifodani hisoblang.
- (isbot) (a)  $\text{ReLU}(z) - \text{ReLU}(-z) = z$  va  $\text{ReLU}(z) + \text{ReLU}(-z) = |z|$  ni isbotlang. (b)  $\max(x_1, x_2) = x_2 + \text{ReLU}(x_1 - x_2)$  dan foydalanib, 3 ta yashirin ReLU neyronli tarmoq ( $W_1, \mathbf{b}_1, W_2, \mathbf{b}_2$ ) quring, u  $\max(x_1, x_2)$  ni aniq hisoblasin;  $(3; -1)$  va  $(-2; -5)$  da tekshiring. (c)  $\text{softplus} \ln(1 + e^z)$  ning hosilasi  $\sigma(z)$  ekanini ko'rsating.
- (hisob) Faollashtirishsiz ikki qatlam:  $\mathbf{h} = W_1 \mathbf{x} + \mathbf{b}_1$ ,  $\mathbf{y} = W_2 \mathbf{h} + \mathbf{b}_2$ , bunda  $W_1 = \begin{pmatrix} 1 & 2 \\ 0 & -1 \\ 3 & 1 \end{pmatrix}$ ,  $\mathbf{b}_1 = (1; 0; -1)$ ,  $W_2 = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \end{pmatrix}$ ,  $\mathbf{b}_2 = (0,5; -1)$ . Unga teng bitta qatlam  $W, \mathbf{b}$  ni toping,  $\mathbf{x} = (1; -1)$  da tekshiring va ikki holatdagi parametrlar sonini solishtiring.
- (hisob, tushuncha) To'rtta skalyar neyron zanjiri  $a_l = \varphi(z_l)$ ,  $z_l = a_{l-1} + b_l$  ( $w_l = 1$ ,  $l = 1, \dots, 4$ ) berilgan; oldindan faollashtirishlar  $z = (0,5; 1; -2; 3)$ .  $\partial a_4 / \partial a_0 = \prod_l \varphi'(z_l)$  ni sigmoid, ReLU va Leaky ReLU ( $\alpha = 0,01$ ) uchun hisoblang. Sigmoid uchun natijani  $0,25^4$  bilan solishtiring. ReLU natijasini izohlang.
- (hisob) Perceptron ( $\hat{y} = \mathbf{1}[\mathbf{w}^T \mathbf{x} + b \geq 0]$ ,  $\eta = 1$ ,  $\mathbf{w}_0 = \mathbf{0}$ ,  $b_0 = 0$ ) quyidagi tartibda:  $(1, 1) \mapsto 0$ ,  $(2, 3) \mapsto 1$ ,  $(3, 1) \mapsto 0$ ,  $(1, 3) \mapsto 1$ . Har bir epoxa oxiridagi xatolar soni,  $\mathbf{w}$ ,  $b$  jadvalini tuzing; algoritim qachon to'xtaydi va jami nechta yangilanish bo'ladi?
- (isbot, hisob) 8-topshiriq ma'lumotlari uchun bias trick ( $\tilde{\mathbf{x}} = (\mathbf{x}; 1)$ ,  $\tilde{y} = 2y - 1$ ) va  $\mathbf{u} = \frac{1}{3}(-2; 2; -1)$  bilan:  $\|\mathbf{u}\| = 1$  ekanini,  $R = \max\|\tilde{\mathbf{x}}\|$ ,  $\gamma = \min \tilde{y} \mathbf{u}^T \tilde{\mathbf{x}}$  va Novikoff chegarasi  $(R/\gamma)^2$  ni toping. Uni 8-topshiriqdagi haqiqiy yangilanishlar soni bilan solishtiring.
- (isbot) Bir o'lchovli ma'lumot  $x = (-2, -1, 0, 1, 2)$ ,  $y = (1, 0, 0, 0, 1)$  chiziqli ajraluvchi emasligini isbotlang.  $\phi(x) = x^2$  belgisi bilan ajratuvchi  $w, b$  ni toping va yangi fazodagi geometrik margin  $\min |z|/|w|$  ni hisoblang. Bu yashirin qatlam g'oyasi bilan qanday bog'liq?
- (hisob) XNOR  $((0, 0) \mapsto 1, (0, 1) \mapsto 0, (1, 0) \mapsto 0, (1, 1) \mapsto 1)$  ni aniq hisoblaydigan 2 ta yashirin ReLU

neyronli va chiziqli chiqishli tarmoq quring ( $W_1, \mathbf{b}_1, \mathbf{w}_2, b_2$  ni yozing) va to'rt nuqta uchun jadval bilan tekshiring.

12. (*hisob*) Logitlar  $\mathbf{z} = (2; -1; 0.5)$ , belgilar  $\mathbf{y} = (1; 0; 0)$ . Har bir namuna uchun binary cross-entropy  $\ell_k = \ln(1 + e^{z_k}) - y_k z_k$ , o'rtacha  $L$  va  $\partial L / \partial z_k$  ni toping. So'ng  $z = -4$ ,  $y = 1$  uchun BCE va MSE ( $\frac{1}{2}(p - y)^2$ ) bo'yicha  $\partial \ell / \partial z$  ni va ularning nisbatini hisoblang.
13. (*hisob*) Logistik neyron,  $\mathbf{x}^{(1)} = (1; 1)$ ,  $y_1 = 1$ ;  $\mathbf{x}^{(2)} = (2; -1)$ ,  $y_2 = 0$ ;  $\mathbf{w} = \mathbf{0}$ ,  $b = 0$ ,  $\eta = 0.5$ ,  $L = \frac{1}{2} \sum_k \ell_k$ . Bitta to'liq (full-batch) gradient qadamini bajaring:  $\nabla_{\mathbf{w}} L$ ,  $\partial L / \partial b$ , yangi  $\mathbf{w}$ ,  $b$  va yangi  $L$  ni toping.
14. (*amaliy holat: tibbiy skrining*) Yurak xastaligi xavfi modeli:  $z = -5 + 0.04 \cdot \text{yosh} + 1.2 \cdot \text{chekish} + 0.03 \cdot (\text{qon bosimi} - 120)$ ,  $p = \sigma(z)$  (chekish  $\in \{0, 1\}$ , qon bosimi mm sim. ust.). (a) 60 yoshli, chekadigan, qon bosimi 140 bo'lgan bemor uchun  $p$  ni toping. (b) Chekish va yoshning 10 yilga ortishi odds ni necha marta oshiradi? (c) 60 yoshli chekmaydigan bemor uchun qon bosimining qaysi qiymatidan boshlab  $p \geq 0.2$  bo'ladi?
15. (*hisob*)  $f(\boldsymbol{\theta}) = \frac{1}{2}(\theta_1^2 + 10\theta_2^2)$ ,  $\boldsymbol{\theta}_0 = (1; 1)$ . Gradient tushishning ikki qadamini  $\eta = 0.1$ ,  $\eta = 0.25$  va  $\eta = 2/11$  uchun bajaring va  $f(\boldsymbol{\theta}_2)$  ni toping. Lipschitz doimiysi  $\beta$  ni va yaqinlashish sharti  $\eta < 2/\beta$  ni yozing; qaysi  $\eta$  da tushish uzoqlashadi va nima uchun?
16. (*hisob, tushuncha*)  $d_{\text{in}} = 784$ ,  $d_{\text{out}} = 256$  qatlam uchun Xavier (Glorot) normal standart chetlanishi, Xavier uniform chegarasi  $\sqrt{6/(d_{\text{in}} + d_{\text{out}})}$  va He (Kaiming) normal standart chetlanishini toping. Barcha og'irliklar nolga (yoki bir xil qiymatga) tenglashtirilsa nima uchun o'qitish buziladi? Ikki yashirin neyron bir xil kiruvchi va chiquvchi og'irliklarga ega bo'lsa, ularning gradientlari ham tengligini ko'rsating.

## B. Dasturlash (17–30)

**Umumiy qoidalar.** `import numpy as np, torch, torch.nn as nn`; har bir PyTorch modeli yaratilishidan oldin `torch.manual_seed(42)`; NumPy tasodifiyligi uchun `np.random.default_rng(42)`; boshida `torch.set_num_threads(1)` va `torch.use_deterministic_algorithms(True)`. Barqaror sigmoid: `np.exp(-np.logaddexp(0, -z))` (ogohlantirishsiz). Kod python -W error::DeprecationWarning solutions.py bilan xatosiz ishlashi kerak. Chop etish formati har topshiriqda ko'rsatilgan.

17. (*faollashtirishlar*) `activations(z, alpha=0.01) -> dict` funksiyasini yozing: kalitlar `sigmoid`, `tanh`, `relu`, `leaky_relu`, `softplus`, qiymatlar  $(\varphi(z), \varphi'(z))$  juftligi ( $\text{ReLU}'(0) = 0$ ). `z = np.linspace(-3, 3, 7)` da natijani PyTorch autograd (`float64`; `torch.sigmoid`, `torch.tanh`, `torch.relu`, `F.leaky_relu(t, 0.01)`, `F.softplus`) bilan `np.allclose(..., atol=1e-12)` orqali solishtiring. Har bir funksiya uchun bitta qator: nomi, `np.round(phi_prime, 4)`, `match: True/False`. Hujjat: <https://pytorch.org/docs/stable/nn.functional.html>.
18. (*chiziqli qatlamlar*) `merge_linear(l1: nn.Linear, l2: nn.Linear) -> nn.Linear` yozing:  $l_2(l_1(\mathbf{x}))$  ga teng bitta qatlam ( $W = W_2 W_1$ ,  $\mathbf{b} = W_2 \mathbf{b}_1 + \mathbf{b}_2$ ). `torch.manual_seed(42)`; `l1 = nn.Linear(4, 3)`; `l2 = nn.Linear(3, 2)`; `x = torch.randn(5, 4)`. Chop eting: yangi qatlam `weight` va `bias` shakllari, parametrlar soni (ikki qatlam va bitta qatlam), `max|diff| < 1e-6: True/False`.
19. (*barqaror BCE*) `bce_naive(z, y) (-y ln p - (1 - y) ln(1 - p), np.errstate(all='ignore'))` ichida va `bce_stable(z, y) = max(z, 0) - yz + ln(1 + e-|z|) (np.log1p)` funksiyalarini yozing. `z = [-800, -30, 0, 30, 800]`, `y = [1, 0, 1, 1, 0]` uchun ikkalasini va `F.binary_cross_entropy_with_logits(..., reduction='none')` ni `f"{v:.4g}"` formatida chop eting; `np.allclose` bilan barqaror shakl va PyTorch mosligini tekshiring.
20. (*perceptron*) `perceptron(X, y, eta=1.0, T=100) -> (w, b, epochs, updates, converged)` (Algoritmi 1,  $\hat{y} = \mathbf{1}[z \geq 0]$ ,  $\mathbf{w}_0 = \mathbf{0}$ ,  $b_0 = 0$ , namunalar ma'lumotdagi tartibda). Ma'lumot: `iris = load_iris()`, faqat `target < 2`, belgilar `data[:, 2:4]` (petal length, petal width),  $y = 1$  — versicolor. Chop eting: `w (np.round(w, 2))`, `b`, epoxalar, yangilanishlar, `converged` va `train accuracy (.3f)`.
21. (*pocket perceptron*) `Iris target > 0`, belgilar `data[:, 2:4]`,  $y = 1$  — virginica. (a) 20-topshiriqdagi perceptron bilan `T=100: converged` va jami yangilanishlar. (b) `pocket_perceptron(X, y, eta=1.0, T=100)`: xuddi shu qoida, lekin *har bir yangilanishdan keyin* `train accuracy` hisoblanib, qat'iy yaxshiroq bo'lsa ( $\mathbf{w}, b$ ) saqlanadi. 100-epoxadagi xatolar sonini hamda eng yaxshi  $\mathbf{w}, b$ , `accuracy (.2f)` ni chop eting. Nega oddiy perceptron to'xtamaydi?
22. (*amaliy holat: ko'krak saratoni diagnostikasi*) `X, y = load_breast_cancer(return_X_y=True)`; `train_test_split(X, y, test_size=0.3, stratify=y, random_state=42)`; `StandardScaler` faqat `train`'da fit qilinadi. `train_logistic(X, y, eta, epochs) -> (w, b, hist)`:  $\mathbf{w} = \mathbf{0}$ ,  $b = 0$ , to'liq gradient tushish, `hist[e]` —  $e$ -yangilanishdan oldingi o'rtacha BCE (`np.logaddexp`), jami `epochs` ta yangilanish va `epochs+1` ta qiymat.  $\eta = 0.5$ , 300 epoxa. Chop eting: `hist` ning 0, 10, 100, 300 elementlari (`.4f`), `train/test accuracy (z ≥ 0 bo'lsa 1)`,  $|w_i|$  bo'yicha eng katta 3 ta belgi nomi va  $w_i$ . Hujjat: [https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load\\_breast\\_cancer.html](https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_breast_cancer.html).
23. (*backprop tekshiruv*) `mip_grads(X, y, W1, b1, w2, b2) -> (L, gW1, gb1, gw2, gb2)` — bir yashirin

ReLU qatlamli tarmoq ( $z = \text{ReLU}(XW_1 + b_1)w_2 + b_2$ , o'rtacha BCE) uchun 1-ma'ruza formulalari bilan qo'lda gradient. Ma'lumot aynan shu tartibda: `rng = np.random.default_rng(42); X = rng.normal(size=(8, 3)); y = rng.integers(0, 2, 8).astype(float); W1 = rng.normal(size=(3, 5)); b1 = rng.normal(size=5); w2 = rng.normal(size=5); b2 = float(rng.normal())`. Natijani float64 autograd (`F.binary_cross_entropy_with_logits`) bilan solishtiring. Chop eting: ikkala yo'qotish (.6f), har bir gradient uchun  $\max|\text{diff}| < 1e-12$ ,  $\partial L / \partial b_2$  (6 xona).

24. (*qadam va  $\beta$* ) 22-topshiriqning standartlangan train to'plamida  $\tilde{X} = [X, \mathbf{1}]$ ,  $\beta = \lambda_{\max}(\tilde{X}^T \tilde{X}) / (4n)$  (`np.linalg.eigvalsh`).  $n$ ,  $\beta$ ,  $2/\beta$  ni chop eting (.4f). So'ng  $\eta \in \{0.01; 0.1; 0.9 \cdot 2/\beta; 5; 50; 500\}$  uchun `train_logistic(..., epochs=100)` bajaring va har biri uchun oxirgi BCE hamda monotone (`np.all(np.diff(hist) <= 1e-12)`) ni chop eting. Nega  $\eta > 2/\beta$  da ham yo'qotish kamayishi mumkin?
25. (*NumPy MLP*) `make_moons(n_samples=400, noise=0.2, random_state=42); train_test_split(..., test_size=0.25, random_state=42)`. 2-16-1 tarmoq, yashirin qatlam tanh, chiqish logit. Boshlang'ich qiymat: `rng = np.random.default_rng(42); W1 = rng.normal(0, 1, (2, 16)) * np.sqrt(1/2); W2 = rng.normal(0, 1, (16, 1)) * np.sqrt(1/16)`; siljishlar nol. Qo'lda backprop, to'liq gradient tushish  $\eta = 0.5$ , qadamlar  $s = 0, 1, \dots, 3000$  (3001 ta yangilanish).  $s \in \{0, 1000, 2000, 3000\}$  da yangilanishdan oldingi BCE (.4f) va oxirida train/test accuracy ni chop eting.
26. (*faollashtirishlarni solishtirish*) `make_circles(n_samples=500, noise=0.1, factor=0.5, random_state=42)`, float32 tenzorlar. Har bir  $\varphi \in \{\text{nn.Sigmoid}, \text{nn.Tanh}, \text{nn.ReLU}\}$  uchun: `torch.manual_seed(42); nn.Sequential(nn.Linear(2, 8), act(), nn.Linear(8, 1)); torch.optim.SGD(lr=0.5, nn.BCEWithLogitsLoss(), 1000` ta to'liq paketli qadam. Oxirgi BCE (.4f) va accuracy (.3f) ni chop eting va farqni izohlang.
27. (*gradientning so'nishi*) `torch.manual_seed(42); x = torch.randn(64, 32); y = (torch.rand(64, 1) > 0.5).float()`. Tarmoq: 20 ta [`nn.Linear(32, 32), act()`] bloki va `nn.Linear(32, 1)`. Ikki variant (har biridan oldin `torch.manual_seed(42)`): (a) `nn.Sigmoid`, standart initialization; (b) `nn.ReLU`, har bir `nn.Linear(32, 32)` yaratilishi bilan `nn.init.kaiming_normal_(weight, nonlinearity='relu')` va nol bias. Bitta `backward()` (BCE with logits) dan keyin birinchi va 20-yashirin qatlam og'irlik gradienti normalarini va ularning nisbatini (.3e) chop eting.
28. (*initialization*) `torch.manual_seed(42); x = torch.randn(1000, 256)`. Uchta usul: `nn.init.normal_(W, 0.0, 0.01)`, `nn.init.xavier_normal_`, `nn.init.kaiming_normal_(W, nonlinearity='relu')`. Har biri uchun `torch.manual_seed(42)`, so'ng 10 marta: `W = torch.empty(256, 256), init(W), h = torch.relu(h @ W.T)`. 1-, 5- va 10-qatlamdan keyingi `h.std()` ni (.3e) chop eting va natijani  $\text{Var } z = dv \mathbb{E}[a^2]$  formulasi bilan tushuntiring.
29. (*universal approksimatsiya*) `x = torch.linspace(-math.pi, math.pi, 200).unsqueeze(1), y = torch.sin(x)`.  $m \in \{2, 8, 32\}$  uchun: `torch.manual_seed(42); nn.Sequential(nn.Linear(1, m), nn.ReLU(), nn.Linear(m, 1)); torch.optim.Adam(lr=0.01)`, 2000 ta to'liq paketli qadam, `F.mse_loss`. Parametrlar soni, oxirgi MSE (.2e) va  $\max|f(x) - \sin x|$  (.4f) ni chop eting. Natija teorema bilan qanday bog'liq?
30. (*amaliy holat: nomutanosib sinflar*) `X, t = load_digits(return_X_y=True); y = (t == 8).float32(), X / 16.0; train_test_split(..., test_size=0.3, stratify=y, random_state=42)`. Logistik neyron `nn.Linear(64, 1)`, `torch.optim.Adam(lr=0.01)`, 300 ta to'liq paketli qadam, ikki variant (har biridan oldin `torch.manual_seed(42)`): `nn.BCEWithLogitsLoss()` va `nn.BCEWithLogitsLoss(pos_weight=n_neg/n_pos)` (train bo'yicha). Train'dagi musbat/manfiy sonlari, `pos_weight` (.3f) va test bo'yicha accuracy, precision, recall, F1-score (`sklearn.metrics`, .4f) ni chop eting. Skrining uchun qaysi variant ma'qul? Hujjat: <https://pytorch.org/docs/stable/generated/torch.nn.BCEWithLogitsLoss.html>.